📘 CSE414 — Complete Concept Guide (Quiz + Mid Exam)

🔷 PART 1: Introduction Section — কী এবং কীভাবে লিখতে হয়

Introduction কী?

> **Introduction** = Research paper এর প্রথম section যেখানে reader কে বলা হয় — কী সমস্যা আছে, কেন সেটা গুরুত্বপূর্ণ, আগে কী হয়েছে, এবং এই **paper** কী করবে।

Introduction হলো paper এর **"দরজা"** — reader এখান থেকেই বোঝে paper টা পড়া দরকার কিনা।

Introduction এর ৫টা অংশ — "CARGO" Structure

এই ৫টা অংশ মনে রাখো:

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C — Context        (বিষয়টা কী, কেন গুরুত্বপূর্ণ)
A — Advancement    (এই বিষয়ে আগে কী কী হয়েছে)
R — Research Gap   (কোথায় কমতি আছে)
G — Goal           (এই paper কী করবে)
O — Organization   (paper টা কীভাবে সাজানো)
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প্রতিটা অংশ বিস্তারিত — **Example** সহ

C — Context (প্রেক্ষাপট)

কী লিখবে: Topic টা globally কতটা important, কতজন affected, কী ধরনের সমস্যা।

Example (Medical — Eye Disease):

> Eye diseases such as diabetic retinopathy, glaucoma, and cataracts are among the leading causes of visual impairment and blindness worldwide. According to the World Health Organization (WHO), approximately 2.2 billion people suffer

> from vision impairment, of which at least 1 billion cases could have been prevented with timely diagnosis and treatment. Early and accurate detection of these conditions is therefore critical to preventing irreversible vision loss.

Example (WBC Classification — Mid exam এর abstract):

> White blood cells (WBCs) play a fundamental role in the human immune system, defending the body against infections and diseases. Accurate classification of WBCs is essential for diagnosing a wide range of medical conditions, including leukemia, anemia, and other hematological disorders. Manual classification by pathologists is time-consuming, error-prone, and subject to inter-observer variability, necessitating the development of automated and reliable classification systems.

A — Advancement (আগে কী হয়েছে)

কী লিখবে: এই বিষয়ে গবেষণা কতটুকু এগিয়েছে — briefly।

Example:

> Recent advancements in deep learning and computer vision have enabled the development of automated medical image analysis systems. Convolutional neural networks (CNNs) have been widely adopted for image classification tasks in medical imaging, demonstrating remarkable performance in disease detection from retinal fundus images, histopathological slides, and microscopic blood smear images.

R — Research Gap (গবেষণার ফাঁক)

কী লিখবে: আগের কাজে কী কী সমস্যা ছিল।

Example:

> However, existing approaches suffer from several limitations. Many models are designed for single-disease classification and cannot generalize across multiple disease categories. Furthermore, most deep learning models lack interpretability, making it difficult for clinicians to trust and adopt these systems in real-world medical practice. The complexity and subtlety of visual features in fundus images also pose significant challenges for standard CNN architectures.

G — Goal (এই paper এর লক্ষ্য)

কী লিখবে: তোমার paper কী propose করছে এবং কী achieve করতে চাইছে।

Example:

> To address these limitations, this study proposes EDDNet30, a novel 30-layer deep convolutional neural network incorporating spatial attention modules and multi-scale fusion techniques for accurate multi-class eye disease classification from fundus images. The proposed model aims to improve diagnostic accuracy while enhancing interpretability through explainable AI techniques including Grad-CAM and LIME.

O — Organization (Paper এর structure)

কী লিখবে: Paper এর বাকি sections কী কী নিয়ে।

Example:

> The remainder of this paper is organized as follows. Section 2 presents a review of related literature on eye disease classification. Section 3 describes the proposed methodology and system architecture. Section 4 presents the experimental results and comparative analysis. Section 5 concludes the paper with future research directions.

✅ Complete Introduction Example (Mid Exam Q1a — WBC Paper)

> White blood cells (WBCs) are a vital component of the human immune system, responsible for protecting the body against pathogens and infections. Accurate and efficient classification of WBCs is crucial for medical professionals to diagnose various diseases, including leukemia, lymphoma, and autoimmune disorders. Traditional manual classification methods are labor-intensive, time-consuming, and prone to human error, particularly in high-volume clinical settings.
>
> Recent advances in deep learning have opened new possibilities for automated medical image analysis. Convolutional neural networks (CNNs) have demonstrated strong performance in image recognition tasks and have been increasingly applied to medical image classification, including blood cell analysis.
>
> However, existing deep learning approaches for WBC classification often rely on limited datasets and standard architectures without domain-specific optimization. Many systems also lack interpretability mechanisms, preventing clinicians from understanding the basis of model predictions, which is critical for clinical adoption.

This study presents an enhanced CNN-based model for WBC classification, incorporating advanced image preprocessing techniques including padding, thresholding, erosion, dilation, and masking to improve feature quality. The model is further evaluated against established transfer learning architectures including Inception V3, MobileNetV2, and DenseNet201, and enhanced with explainable AI methods such as SHAP, LIME, Grad-CAM, and Grad-CAM++ to ensure model transparency and interpretability.

The remainder of this paper is structured as follows. Section 2 reviews related literature. Section 3 outlines the proposed methodology. Section 4 presents experimental results. Section 5 concludes the study.

🔷 PART 2: Literature Review — ৩ ভাবে লেখা

পদ্ধতি ১ — Author Name সরাসরি (1st, 2nd, 3rd Author)

যখন author এর নাম explicitly দেওয়া থাকে, তখন এভাবে লেখো:

Format:

> **[1st Author] et al.** proposed/developed [method] for [problem]. [Result]. However, [limitation].

"et al." use করো যখন author ৩ বা তার বেশি। দুইজন হলে**:** Rahman and Ahmed (2021) একজন হলে**:** Rahman (2021)

Example:

> Ran Ran et al. developed an enhanced convolutional neural network for white blood cell classification using multiple image preprocessing techniques including thresholding, erosion, dilation, and masking. The proposed model achieved a testing accuracy of 99.12% and an F1-score of 99%, outperforming transfer learning models such as Inception V3, MobileNetV2, and DenseNet201. However, the study was conducted using a single dataset and the generalizability of the model across diverse clinical environments remains to be validated.

পদ্ধতি ২ — Reference Number দিয়ে [IEEE Style]

Academic paper এ সবচেয়ে common। Reference list এ paper গুলো numbered থাকে, আর text এ সেই number ব্যবহার করো।

Format:

> In [1], the authors proposed [method] for [problem]. [Result]. However, [limitation].

অথবা:

> [method] was proposed in [2] to address [problem], achieving [result]. Despite this, [limitation].

Example:

> In [1], a CNN-based approach was proposed for automated WBC classification using preprocessing techniques such as erosion, dilation, and adaptive masking. The model achieved 99.12% accuracy and was validated against three transfer learning models. However, the study lacked evaluation on cross-institutional datasets, limiting its clinical applicability.
>
> The authors in [2] applied transfer learning using ResNet50 for retinal disease classification and reported an accuracy of 92.6%. Despite its strong baseline performance, the model relied heavily on pretrained ImageNet weights and did not incorporate attention mechanisms for retinal-specific feature extraction.

পদ্ধতি ৩ — Mixed Style (Author Name + Reference Number)

এটা সবচেয়ে professional এবং commonly expected academic style।

Format:

> Smith et al. [1] proposed [method] for [problem]. [Result]. However, [limitation].

Example:

> Gulshan et al. [1] developed a deep learning system trained on a large fundus image dataset for diabetic retinopathy detection, achieving high diagnostic sensitivity. However, the system was limited to binary classification and could not distinguish among multiple disease categories simultaneously.
>
> He et al. [2] introduced ResNet with residual connections, which was widely adapted for medical image classification tasks. While ResNet50 demonstrated competitive accuracy in eye disease detection, its purely convolutional design restricted the modeling of global spatial dependencies across fundus images.

Dosovitskiy et al. [3] proposed the Vision Transformer (ViT), leveraging self-attention mechanisms for global feature modeling. Although ViT-B16 outperformed CNNs on large datasets, it required extensive pretraining data and showed limited performance on smaller domain-specific medical datasets.

✅ Full Literature Review Example — Quiz 2 (Skin Cancer / Melanoma)

Abstract থেকে key info:

- Skin cancer / Melanoma detection
- Deep learning + Machine learning combined
- Contourlet Transform + Local Binary Pattern Histogram
- Kaggle / ISIC Archive dataset
- Ensemble model
- 93% accuracy, 99.7% recall (benign), 86% recall (malignant)

Literature Review (Related Works)

Skin cancer, particularly melanoma, is one of the most fatal forms of cancer, and its automated detection has been a growing focus of research in medical image analysis. Various machine learning and deep learning approaches have been proposed to improve diagnostic accuracy.

Esteva et al. [1] proposed a deep CNN for skin lesion classification trained on a large dermatologist-labeled dataset. The model achieved dermatologist-level accuracy in binary classification of malignant and benign lesions. However, the model relied solely on deep features and did not incorporate handcrafted texture descriptors, limiting its ability to capture subtle lesion patterns.

Codella et al. [2] developed an ensemble method combining deep learning with traditional feature extraction techniques for melanoma detection using the ISIC Archive dataset. While the approach demonstrated improved sensitivity, the system required high computational resources and lacked generalization across diverse skin tone datasets.

Kawahara and Hamarneh [3] applied multi-resolution CNN architectures for dermoscopic image analysis. The model captured features at multiple scales but depended heavily on large annotated training datasets, which are often scarce in clinical environments.

Harangi [4] explored the use of transfer learning models including VGG16 and ResNet for skin lesion classification. While transfer learning reduced training time significantly, these models were not optimized for skin-specific texture features and showed lower recall rates for malignant cases, which is clinically critical.

Despite these advancements, existing methods demonstrate key limitations including insufficient integration of manual and automated feature extraction, low recall rates for malignant cases, and limited interpretability. The proposed ensemble model addresses these gaps by combining deep learning-based automatic feature extraction with handcrafted features derived from Contourlet Transform and Local Binary Pattern Histogram, achieving 93% overall accuracy with a recall of 99.7% for benign and 86% for malignant cases on the Kaggle dataset derived from the ISIC Archive.

🔷 PART 3: Methodology — Diagram + Theory একসাথে

Methodology Theory কী লিখতে হয়?

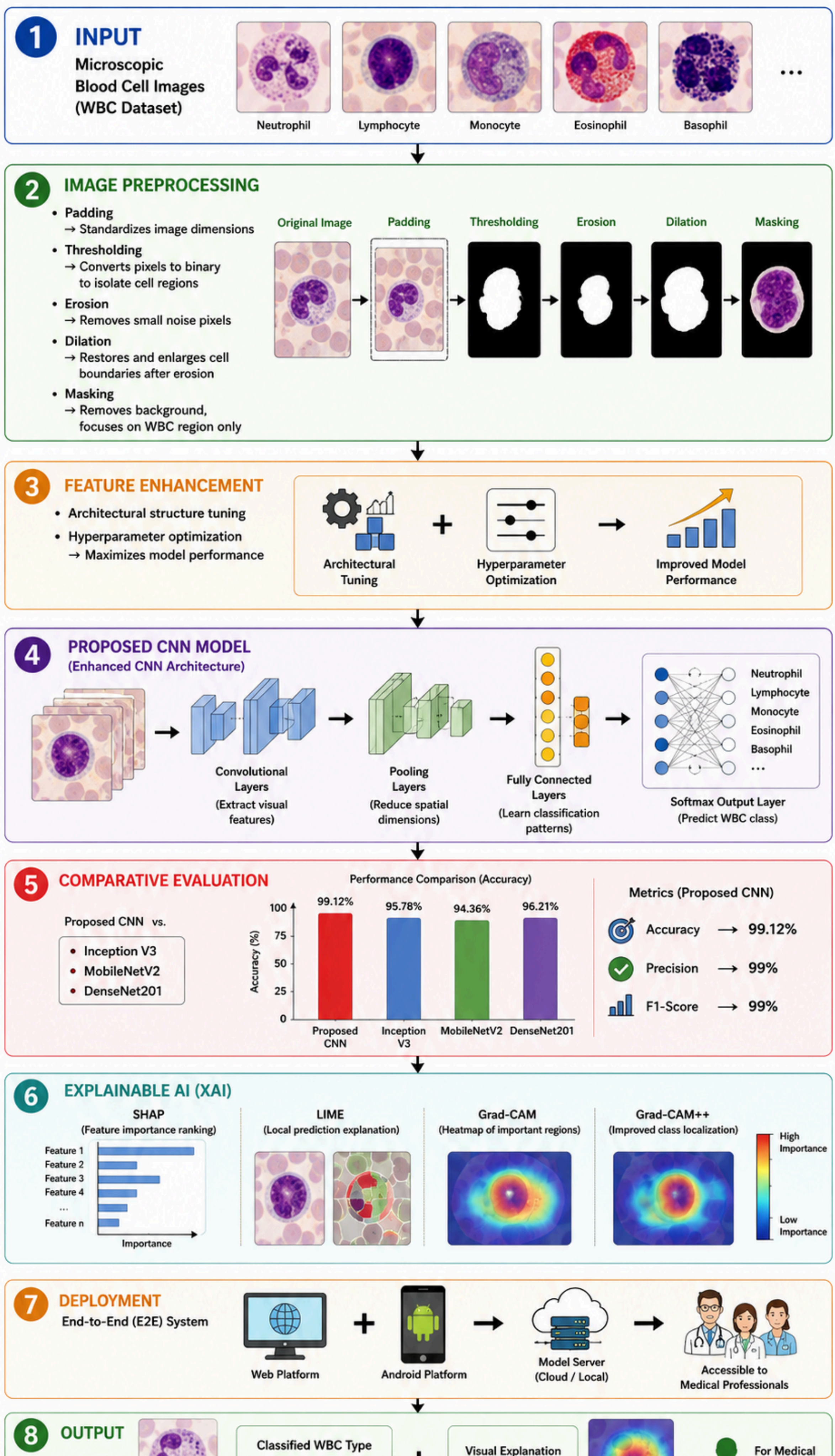
Diagram এর পাশাপাশি প্রতিটা step এর জন্য ১-৩ লাইনের **explanation** লিখতে হয়।

Format:

Step এর নাম → কী করা হয় → কেন করা হয়

✅ Complete Methodology — Mid Exam WBC Paper

Workflow Diagram:


1 INPUT
Microscopic Blood Cell Images (WBC Dataset)
Neutrophil
Lymphocyte
Monocyte
Eosinophil
Basophil
2 IMAGE PREPROCESSING
• Padding → Standardizes image dimensions
• Thresholding → Converts pixels to binary to isolate cell regions
• Erosion → Removes small noise pixels
• Dilation → Restores and enlarges cell boundaries after erosion
• Masking → Removes background, focuses on WBC region only
Original Image
Padding
Thresholding
Erosion
Dilation
Masking
3 FEATURE ENHANCEMENT
• Architectural structure tuning
• Hyperparameter optimization → Maximizes model performance
Architectural Tuning
Hyperparameter Optimization
Improved Model Performance
4 PROPOSED CNN MODEL
(Enhanced CNN Architecture)
Convolutional Layers (Extract visual features)
Pooling Layers (Reduce spatial dimensions)
Fully Connected Layers (Learn classification patterns)
Neutrophil
Lymphocyte
Monocyte
Eosinophil
Basophil
Softmax Output Layer (Predict WBC class)
5 COMPARATIVE EVALUATION
Proposed CNN vs.
• Inception V3
• MobileNetV2
• DenseNet201
Performance Comparison (Accuracy)
Accuracy (%)
100
75
50
25
0
99.12%
95.78%
94.36%
96.21%
Proposed CNN
Inception V3
MobileNetV2
DenseNet201
Metrics (Proposed CNN)
Accuracy → 99.12%
Precision → 99%
F1-Score → 99%
6 EXPLAINABLE AI (XAI)
SHAP (Feature importance ranking)
Feature 1
Feature 2
Feature 3
Feature 4
...
Feature n
Importance
LIME (Local prediction explanation)
Grad-CAM (Heatmap of important regions)
Grad-CAM++ (Improved class localization)
High Importance
Low Importance
7 DEPLOYMENT
End-to-End (E2E) System
Web Platform
Android Platform
Model Server (Cloud / Local)
Accessible to Medical Professionals
8 OUTPUT
Classified WBC Type
Visual Explanation
For Medical

Methodology Theory (প্রতিটা step এর explanation):

Step 1 — Data Collection: A microscopic blood cell image dataset was utilized for WBC classification. The dataset comprises labeled images of different WBC types collected for training and evaluation purposes.

Step 2 — Image Preprocessing: Several preprocessing techniques were applied to enhance image quality and reduce noise. Padding was used to standardize image dimensions. Thresholding converted pixel intensities to binary values to isolate cell regions. Erosion and dilation were applied to reduce noise and refine cell boundary structures respectively. Masking was employed to remove background regions and focus the model on relevant cellular features.

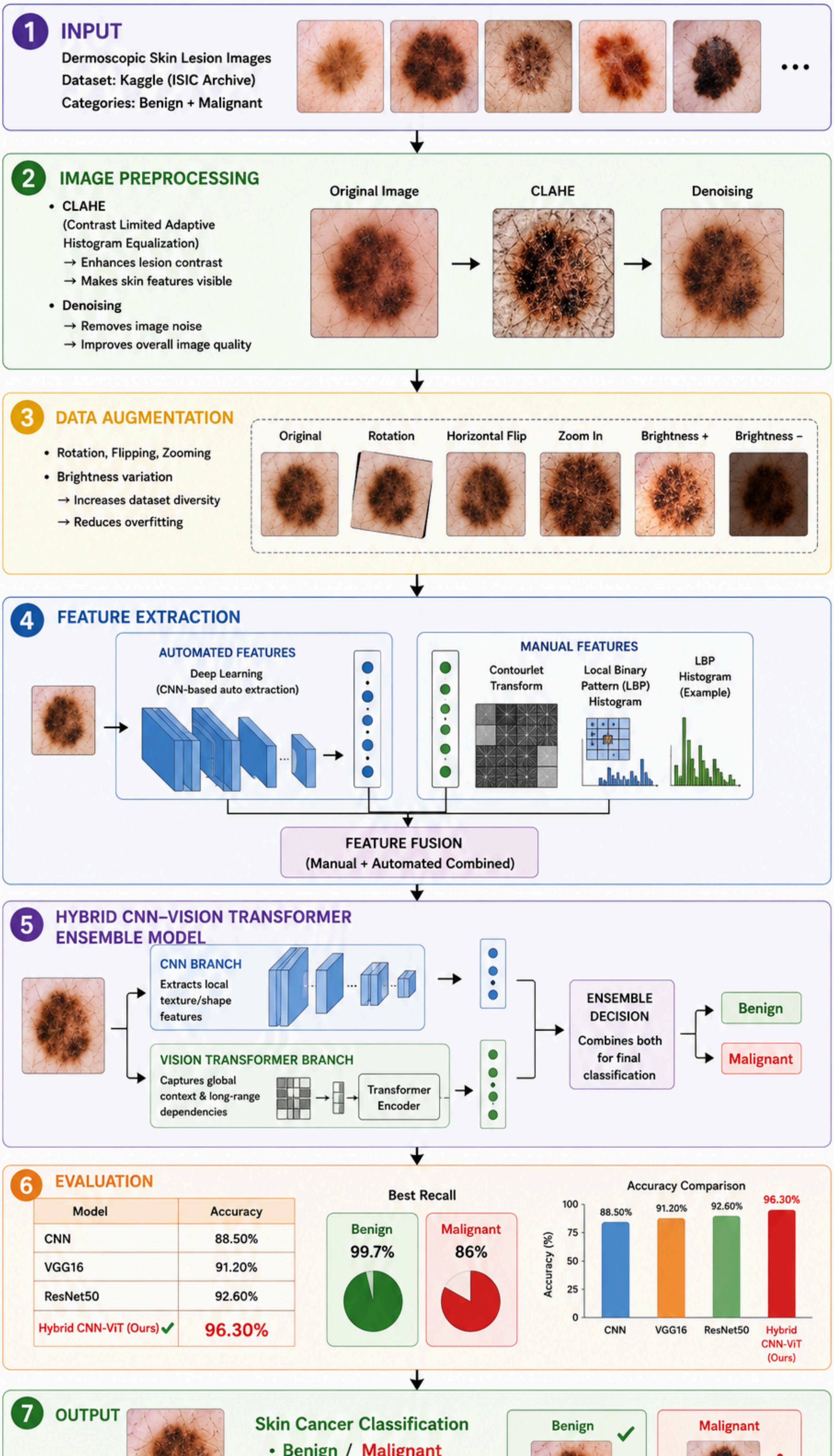
Step 3 — Model Architecture: An enhanced CNN architecture was developed with optimized layers and hyperparameters. The model consists of convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. Architectural structures and hyperparameters were experimented with extensively to maximize classification performance.

Step 4 — Training and Comparison: The proposed model was trained and evaluated against three transfer learning baselines — Inception V3, MobileNetV2, and DenseNet201 — to benchmark its performance. Standard evaluation metrics including accuracy, precision, and F1-score were computed.

Step 5 — Explainable AI: To improve model interpretability, four XAI techniques were applied. SHAP (Shapley Additive Explanations) identified the contribution of each feature to the model's prediction. LIME (Local Interpretable Model-agnostic Explanations) provided local explanations for individual predictions. Grad-CAM and Grad-CAM++ generated visual heatmaps highlighting the regions of the input image most influential in the model's decision-making process.

Step 6 — Deployment: The final model was integrated into an end-to-end (E2E) system accessible via both web and Android platforms, enabling real-time WBC classification for clinical use.

✅ Methodology — Quiz 2 (Skin Cancer / Melanoma)



1 INPUT
Dermoscopic Skin Lesion Images
Dataset: Kaggle (ISIC Archive)
Categories: Benign + Malignant
2 IMAGE PREPROCESSING
• CLAHE
(Contrast Limited Adaptive Histogram Equalization)
→ Enhances lesion contrast
→ Makes skin features visible
• Denoising
→ Removes image noise
→ Improves overall image quality
Original Image
CLAHE
Denoising
3 DATA AUGMENTATION
• Rotation, Flipping, Zooming
• Brightness variation
→ Increases dataset diversity
→ Reduces overfitting
Original
Rotation
Horizontal Flip
Zoom In
Brightness +
Brightness –
4 FEATURE EXTRACTION
AUTOMATED FEATURES
Deep Learning
(CNN-based auto extraction)
MANUAL FEATURES
Contourlet Transform
Local Binary Pattern (LBP) Histogram
LBP Histogram (Example)
FEATURE FUSION
(Manual + Automated Combined)
5 HYBRID CNN–VISION TRANSFORMER ENSEMBLE MODEL
CNN BRANCH
Extracts local texture/shape features
VISION TRANSFORMER BRANCH
Captures global context & long-range dependencies
Transformer Encoder
ENSEMBLE DECISION
Combines both for final classification
Benign
Malignant
6 EVALUATION
Model
Accuracy
CNN
88.50%
VGG16
91.20%
ResNet50
92.60%
Hybrid CNN-ViT (Ours)
96.30%
Best Recall
Benign
99.7%
Malignant
86%
Accuracy Comparison
Accuracy (%)
100
75
50
25
0
88.50%
91.20%
92.60%
96.30%
CNN
VGG16
ResNet50
Hybrid CNN-ViT (Ours)
7 OUTPUT
Skin Cancer Classification
• Benign / Malignant
Benign
Malignant

🔷 PART 4: Quick Reference — Exam এ সব একসাথে

Introduction এর ৫ অংশ মনে রাখার Trick:

C → Context → "এই বিষয়টা কতটা গুরুত্বপূর্ণ"
A → Advance → "আগে কী কী হয়েছে"
R → Research Gap → "কী কী সমস্যা এখনো আছে"
G → Goal → "এই paper কী করবে"
O → Organization → "paper কীভাবে সাজানো"

Literature Review এর ৩ style:

Style 1 → Rahman et al. proposed...
Style 2 → In [1], the authors proposed...
Style 3 → Rahman et al. [1] proposed... ← সবচেয়ে professional

Methodology Diagram এর Universal Flow:

Input → Preprocessing → Augmentation →
Feature Extraction → Model → Evaluation → XAI → Output

Key Terms — Easy Definition:

Term	Easy Definition
CLAHE	Image এর contrast বাড়ানোর advanced technique
Grad-CAM	Model কোন জায়গা দেখে decision নিয়েছে তার heatmap
SHAP	প্রতিটা feature কতটা prediction এ contribute করেছে
LIME	একটা single prediction কেন হলো তার local explanation
Overfitting	Model training data তে ভালো কিন্তু new data তে খারাপ
Augmentation	Data কৃত্রিমভাবে বাড়ানো (rotate, flip, zoom)
Ensemble	একাধিক model combine করে decision নেওয়া

Transfer Learning	আগে trained model অন্য কাজে reuse করা
Cross-validation	Dataset কে ভাগ করে বারবার test করা
XAI	AI model এর decision কেন হলো সেটা explain করা
